Chapter 1

Taylor's Theorem

This chapter is motivated by the question: how can we approximate a function with other, simpler functions, locally? In particular, the approximation of functions by *polynomials*. Polynomials are the natural first choice as they are, in some way, the simplest type of function on $\mathbb{R}$ and have very nice properties.

When discussing Taylor's theorem, we are fundamentally talking about *polynomials*, not power series. A series that defines a function is not just a single series; for every x, you have a different limit. Establishing whether such a series converges, is continuous, or is differentiable requires significant machinery in general, but for functions with a finite Taylor polynomial, these properties are automatic because of the algebraic properties of polynomials.

1.1 Taylor Polynomials

Definition 1.1 (Taylor Polynomial)**.** Let f be n times differentiable at a point a. The n-th order *Taylor polynomial* of f centred at a is:

$$P_{n,a}(x) := \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k.$$

Remark. We can construct a polynomial with prescribed derivatives at a point a by writing it in the "centred" form $\sum_{k=0}^{n} c_k(x-a)^k$. The derivatives of this polynomial at a directly determine the coefficients $c_k = f^{(k)}(a)/k!$.

Geometrically, by forcing the polynomial and the function to share the exact same value, slope, concavity, and higher-order rates of change at a single point a, we are essentially locking their *local geometries* together. The higher the order n, the more rigid this lock becomes, widening the neighbourhood where the polynomial stays glued to the function before inevitably peeling away.

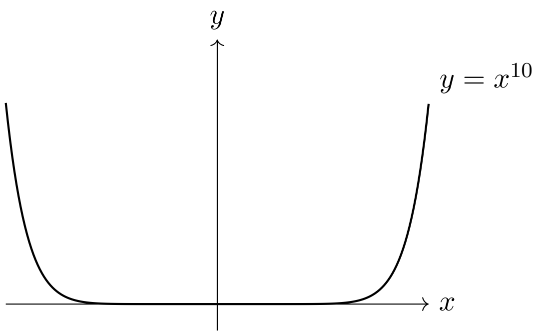
It may seem slightly unnatural to construct a polynomial whose derivatives up to order n to act as a local approximation to a given function, but to see why this works it is useful to consider the *remainder* function for a given function and its Taylor polynomial:

Definition 1.2 (The Remainder)**.** The *remainder* of the Taylor approximation of order n is the difference between the function and its Taylor polynomial:

$$R_{n,a}(x) := f(x) - P_{n,a}(x).$$

The remainder $R_{n,a}(x)$ represents the exact "error" in our approximation. By construction, the function and the polynomial share their first n derivatives at a. Therefore, the remainder function has its first n derivatives vanishing at a.

Geometrically, this corresponds to extreme "flatness" at a (similar to how $y = x^n$ is extremely flat near the origin, for n large - see below). Because the error is practically zero in this flat region, the polynomial "hugs" the function tightly.

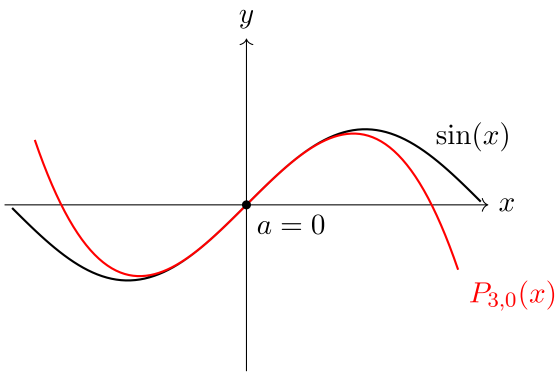


Definition 1.3 (Little-o Notation)**.** Let $k \in \mathbb{N}$. We say that $f \in o(|x-a|^k)$ if

$$\lim_{x \to a} \frac{f(x)}{|x-a|^k} = 0.$$

Conceptually, the extreme flatness of the remainder is demonstrated in the above Little-o notation. As we will show below, $R_{n,a} \in o(|x-a|^n)$.

This limit explicitly states that as x approaches a, the error $R_{n,a}(x)$ shrinks to zero strictly *faster* than the n-th power of the distance from the centre. The error is strictly dominated by the polynomial $|x-a|^n$. This is the formal mathematical translation of the polynomial "hugging" the function: the approximation error vanishes so rapidly that it is negligible even when compared to the microscopic scale of $|x-a|^n$.



Theorem 1.4 (Taylor's Theorem with Lagrange Remainder)**.** *Let f be $n+1$ times differentiable on an open interval containing a and x. Then there exists some θ strictly between a and x such that:*

$$R_{n,a}(x) = \frac{f^{(n+1)}(\theta)}{(n+1)!}(x-a)^{n+1}.$$

Proof. Fix $x \neq a$. Consider the function $F(t)$ defined on the closed interval between a and x:

$$F(t) \coloneqq f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(t)}{k!}(x-t)^k.$$

Notice that $F(x) = 0$ and $F(a) = R_{n,a}(x)$. Differentiating $F(t)$ with respect to t causes the sum to telescope, leaving only the final term:

$$F'(t) = -\frac{f^{(n+1)}(t)}{n!}(x-t)^n.$$

Now, define an auxiliary function $G(t)$ to apply Rolle's Theorem:

$$G(t) \coloneqq F(t) - \left(\frac{x-t}{x-a}\right)^{n+1} F(a).$$

We have $G(x) = 0$ and $G(a) = F(a) - F(a) = 0$. Since G is continuous on the closed interval between a and x and differentiable on the open interval, Rolle's Theorem guarantees a point θ strictly between a and x where $G'(\theta) = 0$.

Differentiating $G(t)$ gives:

$$G'(t) = F'(t) + (n+1)\frac{(x-t)^n}{(x-a)^{n+1}}F(a).$$

Setting $G'(\theta) = 0$ and substituting $F'(\theta)$ yields:

$$\frac{f^{(n+1)}(\theta)}{n!}(x-\theta)^n = (n+1)\frac{(x-\theta)^n}{(x-a)^{n+1}}R_{n,a}(x).$$

Cancelling $(x-\theta)^n$ (since $\theta \neq x$) and rearranging gives the desired form for $R_{n,a}(x)$. $\square$

Remark. The Lagrange form is incredibly convenient for rough approximations. If the $(n+1)$-th derivative is bounded, the remainder is immediately bounded. However, it has significant drawbacks: the value θ depends on x in a way that is not explicitly understood and is generally not continuous. Therefore, if we want to differentiate the remainder or evaluate it precisely, the Lagrange form is inadequate.

Theorem 1.5 (Taylor's Theorem with Integral Remainder)**.** *Let f be $n+1$ times continuously differentiable on an open interval containing a. For any x in this interval,*

$$R_{n,a}(x) = \int_a^x \frac{f^{(n+1)}(t)}{n!}(x-t)^n \,\mathrm{d}t.$$

Proof. We proceed by mathematical induction on n. For $n = 0$, the theorem states $f(x) - f(a) = \int_a^x f'(t)\,\mathrm{d}t$, which is exactly the Fundamental Theorem of Calculus.

Assume the formula holds for some k. We apply integration by parts to the integral remainder for k. Let $u = f^{(k+1)}(t)$ and $v = -\frac{(x-t)^{k+1}}{k+1}$. Then $\mathrm{d}u = f^{(k+2)}(t)\,\mathrm{d}t$ and $\mathrm{d}v = (x-t)^k\,\mathrm{d}t$.

$$\int_a^x \frac{f^{(k+1)}(t)}{k!}(x-t)^k\,\mathrm{d}t = \left[-f^{(k+1)}(t)\frac{(x-t)^{k+1}}{(k+1)!}\right]_a^x + \int_a^x \frac{f^{(k+2)}(t)}{(k+1)!}(x-t)^{k+1}\,\mathrm{d}t.$$

Evaluating the boundary term at x yields 0. Evaluating at a yields $\frac{f^{(k+1)}(a)}{(k+1)!}(x-a)^{k+1}$. Thus,

$$R_{k,a}(x) = \frac{f^{(k+1)}(a)}{(k+1)!}(x-a)^{k+1} + \int_a^x \frac{f^{(k+2)}(t)}{(k+1)!}(x-t)^{k+1}\,\mathrm{d}t.$$

Since $R_{k,a}(x) - \frac{f^{(k+1)}(a)}{(k+1)!}(x-a)^{k+1} = R_{k+1,a}(x)$, the result holds for $n = k+1$. $\square$

Remark. The integral remainder provides an *explicit* formula for the remainder that works for every x. Because the dependence on x is explicitly captured inside the integral (and not hidden in an arbitrary θ), we can use the Fundamental Theorem of Calculus to differentiate the remainder, bound it precisely, and evaluate it anywhere.

1.2 Taylor Series

We have seen that Taylor polynomials are fundamentally local approximations, however there is a special class of functions, called *analytic functions*, whose Taylor polynomials converge to the original function on an interval as $n \to \infty$. This is quite surprising, as it says that these functions can be recovered *globally* in the form of an "infinite polynomial" from only their derivatives *at a single point*, i.e. from only local information. Indeed, it can feel deeply counter-intuitive that purely local information - derivatives evaluated at a single microscopic point a - can perfectly reconstruct a function's global shape. If you draw an arbitrary curve by hand, knowing its slope at $x = 0$ tells you absolutely nothing about its height at $x = 10$.

However, polynomials are extremely rigid. A polynomial of degree n cannot suddenly sprout a new wiggle or change its asymptotic behaviour; its entire fate across the whole real line is permanently sealed by its finite coefficients. In other words, its global behaviour is completely determined by a finite set of points. Because an analytic function is essentially an "infinite polynomial", it inherits this extreme global rigidity. If you know a real analytic function and all of its derivatives at a single point, you possess the entire "genetic code" of the function. The Taylor coefficients lock the *local* geometry in place, propagating that local curvature outward. There is zero room for the function to do anything unexpected globally because its global behaviour is a rigid algebraic consequence of its local derivatives.

Indeed, these *are* the special functions defined by this property; there are many examples, as we will see below, of functions that are in C^∞ (infinitely differentiable) but are not analytic.

Definition 1.6 (Real Analytic Function)**.** We say that a function $f \in C^\infty$ is *real analytic* at a point a if there exists $R > 0$ such that for every $x \in (a - R, a + R)$ we have $\lim_{n\to\infty} P_{n,a}(x) = f(x)$. In this case, we write

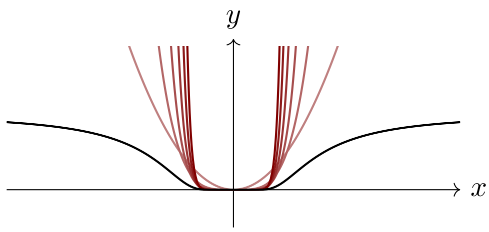
$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

The maximal $R > 0$ for which this property holds is called the *radius of convergence* of the Taylor series.

Intuitively, this means that the function is "rigid" enough that it can be approached globally (i.e. on an interval) by polynomial functions. This is precisely why the derivatives of the function at one point generate can build the whole function; as we saw in the discussion at the start of this chapter, polynomials are extremely special functions that can be re-centred to any point while remaining invariant, and so the local information of an analytic function inherits this property, such that local information, like derivatives at a point a, can still recover the function globally.

Example 1.7. Not all functions in C^∞ are (real) analytic. A classic counterexample is the bump function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} \exp(-1/x^2), & x \neq 0, \\ 0, & x = 0. \end{cases}$$



This function is infinitely differentiable everywhere (which one should verify), but $f^{(n)}(0) = 0$ for all $n \in \mathbb{N}$. Therefore, its Taylor polynomial at $a = 0$ is the identically zero function, $P_{n,0}(x) = 0$ for all n.

Intuitively, becauase this function converges to zero as $x \to 0$ faster than any polynomial, each power x^n is "too weak" to match the exponential decay of f near 0, so has nothing to contribute to the Taylor polynomial at 0. Precisely, it is because $f \in o(|x|^n)$ for every $n \in \mathbb{N}$; the function itself vanishes faster than any polynomial term x^n can measure, so its best polynomial approximation at the origin is simply the zero polynomial, $P_{n,0}(x) = 0$. The polynomials literally cannot "see" the function's global growth because it is locally too flat.

Based on the Lagrange form of the remainder in Taylor's Theorem, we get a useful sufficient condition for the convergence of Taylor series to the original function.

Theorem 1.8. *Let $f \in C^\infty$ be defined on $(a - \rho, a + \rho)$ for some $\rho > 0$. If there exists $C, R > 0$ and $N \in \mathbb{N}$ such that for all $x \in (a - \rho, a + \rho)$ and all $n \geq N$,*

$$|f^{(n+1)}(x)| \leq C \frac{(n+1)!}{R^{n+1}},$$

then the Taylor polynomial $P_{n,a}(x)$ of f converges to $f(x)$ as $n \to \infty$ for every x satisfying

$$|x-a| < \min\{\rho, R\}.$$

In particular, f is analytic on $|x-a| < \min\{\rho, R\}$.

Proof. Using the Lagrange form of the remainder (Theorem 1.4), we have, for x such that $|x-a| < \min\{\rho, R\}$,

$$R_{n,a}(x) = \frac{f^{(n+1)}(\theta)}{(n+1)!}(x-a)^{n+1},$$

for some θ between a and x. Since $|x-a| < \min\{\rho, R\}$, and θ is between a and x, we also have $|\theta - a| < \min\{\rho, R\}$, so for $n \geq N$, the derivative bound applies at θ also:

$$|R_{n,a}(x)| = \left|\frac{f^{(n+1)}(\theta)}{(n+1)!}(x-a)^{n+1}\right| \leq \frac{C\frac{(n+1)!}{R^{n+1}}}{(n+1)!}|x-a|^{n+1} = C\left(\frac{|x-a|}{R}\right)^{n+1}.$$

Since, in particular, $|x-a| < R$ we know $\frac{|x-a|}{R} < 1$. Therefore, as $n \to \infty$, the remainder $|R_{n,a}(x)| \to 0$. This proves that the sequence of Taylor polynomials converges pointwise to $f(x)$. □

Remark. The expression $\frac{(n+1)!}{R^{n+1}}$ is highly advantageous when bounding error because factorials grow significantly faster than exponentials. This affords us a massive amount of "room" to bound the derivatives of functions like $\sin(x)$, $\cos(x)$, and $\exp(x)$, which are indeed analytic functions.

The distinction between infinitely differentiable and analytic is better understood via Complex Analysis (the analysis of $\mathbb{C}$).

In $\mathbb{R}$, being infinitely differentiable is a surprisingly weak geometric condition. Because you can only approach a point from two directions (left or right) on the 1D real line, it is easy to "cheat" and smoothly glue a completely flat line to a curved function (like our bump function) without breaking differentiability.

However, in the complex plane, a function takes inputs from a 2D surface. To be complex-differentiable at a point, the limit of the difference quotient must perfectly agree no matter which of the infinite possible directions you approach from. This geometric constraint is so incredibly rigid that if a complex function is differentiable just *once* on an open set, it is automatically infinitely differentiable and perfectly equal to its Taylor series everywhere in that domain. In the complex world, the "bump function" is mathematically impossible.

Exercise. Prove that $\exp(x)$ is real analytic.